

Exercise 1 - Sagelva

In this exercise we will find the catchment area for the Svarttjønnbekken gauging station in the Sagelva research catchment close to Jonsvatnet in Trondheim. This is typically the first step in a hydrological analysis. We will then calculate some statistics and create some plots that describe the flow regime we measure at Svarttjønnbekken. This kind of analysis is done in many hydrological projects to get an overview of the discharge regime at the station – we will use these data later in the course for further analysis

Task 1 - Catchment

- *Find the catchment area for Svarttjønnbekken gauging station. The position is marked on the map in this exercise with a red circle. There is a map you can print in A3 format (easiest to work with) on Blackboard, or some copies are available in the cupboard outside my office.*

The first step in the hydrological analysis was to delineate the catchment upstream of the **Svarttjønnbekken gauging station**. The station is indicated by the red circle on the topographic map provided for the exercise. The map has a scale of **1:17,000**, allowing the extent of the drainage basin to be determined from the local topography.

The catchment boundary was delineated by identifying the **topographic drainage divides** surrounding the gauging station. These divides correspond to the areas of higher ground separating Svarttjønnbekken from neighbouring drainage basins. The boundary was therefore traced along the local ridgelines, following the principle that surface runoff is directed downslope towards the drainage network and ultimately towards the gauging station.

Based on the delineated drainage basin and the map scale, the catchment area was estimated as:

$A \approx 3.04 \text{ km}^2$

This value is consistent with published information for the Svarttjønnbekken gauging station. The catchment extends from approximately **280 to 513 m a.s.l.**, giving a total topographic relief of approximately **233 m**.

The catchment area is a fundamental parameter for the subsequent hydrological analysis, as it allows discharge measurements to be converted into an equivalent runoff

depth. This is particularly important for the estimation of the annual water balance carried out in Task 2.

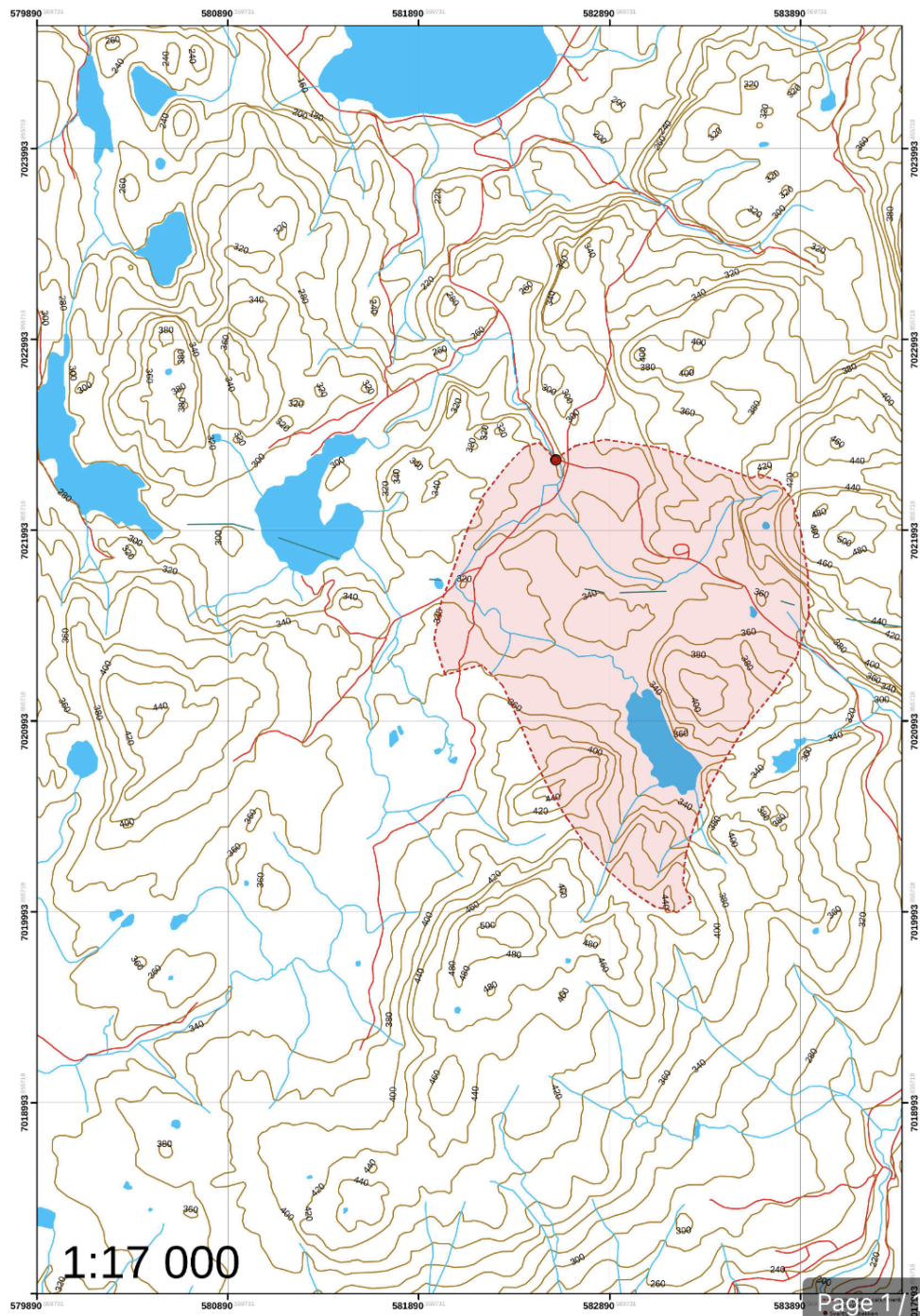


Figure 1. Delineation of the Svarttjønnbekken catchment upstream of the gauging station. The watershed boundary was interpreted from the topographic drainage divides shown

- Find the hypsographic curve (elevation distribution) for the Svarttjønnbekken catchment. Use 100-meter intervals in this computation.

The elevation distribution of the Svarttjønnbekken catchment was determined using **100 m elevation intervals**, as specified in the exercise. A hypsographic curve represents the relationship between the elevation of a drainage basin and the proportion of its total area occurring at or below a given elevation.

The catchment extends from approximately **280 to 513 m a.s.l.** The estimated distribution of the catchment area across the elevation intervals is presented in Table X.

Elevation interval (m a.s.l.)	Area (km ²)	Area (%)
280–380	1.80	59.2
380–480	1.00	32.9
480–513	0.24	7.9
Total	3.04	100.0

The corresponding cumulative distribution was used to construct the hypsographic curve (Figure 2).

Elevation (m a.s.l.)	Cumulative area (km ²)	Cumulative area (%)
280	0.00	0.0
380	1.80	59.2
480	2.80	92.1
513	3.04	100.0

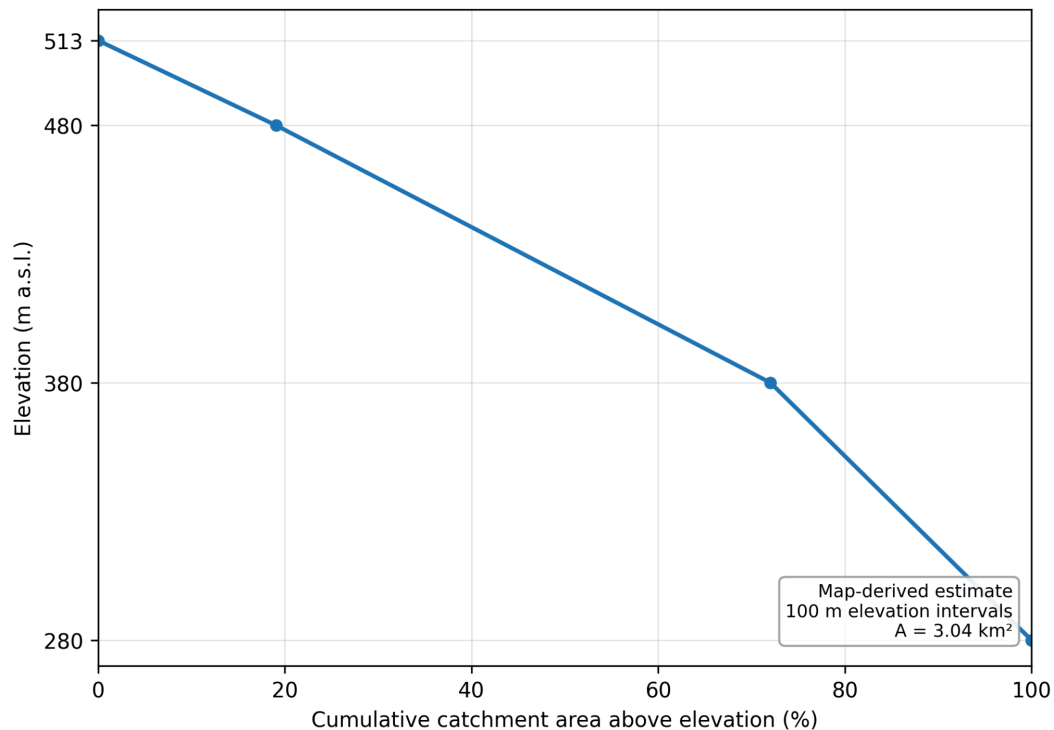


Figure 2. Hypsographic curve of the Svarttjønnbekken catchment based on 100 m elevation intervals. The elevation–area relationship represents a map-derived estimate from the 1:17,000 topographic map.

The hypsographic distribution indicates that the Svarttjønnbekken catchment is predominantly characterised by **low- to intermediate-elevation terrain**. Approximately **59% of the catchment lies below 380 m a.s.l.**, while a further **33% occurs between 380 and 480 m a.s.l.** Only approximately **8% of the catchment is situated above 480 m a.s.l.**

The catchment therefore has a relatively concentrated elevation distribution, despite having a total relief of approximately **233 m**. The higher parts of the basin may contribute to seasonal snow accumulation and subsequent snowmelt, whereas the lower portions are more directly connected to the gauging station and the downstream drainage response.

The elevation distribution is consequently relevant to the interpretation of the discharge regime analysed in Task 2, particularly because seasonal variations in precipitation and snowmelt can influence the timing and magnitude of streamflow.

Task 2 - discharge data

Daily discharge data from the Svarttjønnbekken gauging station were analysed for the period **1973–2025**. The dataset contains daily discharge measurements expressed in m^3/s , with each year stored as a separate column. The calculations and graphical analysis were performed using Python, following the methodology introduced in the exercise.

For each year, the arithmetic mean of the available daily discharge measurements was calculated to characterise the annual flow regime. In addition, the mean discharge for the entire observation period was determined. The daily dataset was subsequently analysed statistically using the minimum, 25th percentile, median, 75th percentile and maximum discharge. Finally, annual maximum discharges were extracted and linear regressions were used to assess possible long-term trends.

- *Find the average discharge for each year and the average discharge for the whole period. Plot the first in a graph to see the variation. What is the average annual runoff volume?*

The calculated annual mean discharge shows considerable interannual variability throughout the observation period (Figure X). The mean discharge for the complete 1973–2025 dataset was:

$$\bar{Q}=0.0997 \text{ m}^3 \text{ s}^{-1}$$

The highest annual mean discharge occurred in **1997**, with a value of **$0.1618 \text{ m}^3/\text{s}$** , whereas the lowest occurred in **2014**, with **$0.0459 \text{ m}^3/\text{s}$** . This variation reflects the substantial year-to-year variability characteristic of a small catchment, where changes in precipitation, snow accumulation and melt, and seasonal hydrological conditions can strongly influence streamflow.

The average annual runoff volume was estimated from the long-term mean discharge according to:

$$V=\bar{Q}\times t$$

where t represents the number of seconds in one year. Using 365.25 days per year gives:

$$V=0.0997\times 365.25\times 24\times 3600$$

and therefore:

$$V=3.15 \times 10^6 \text{ m}^3 \text{ yr}^{-1}$$

Thus, the Svarttjønnbekken catchment produces an average runoff volume of approximately **3.15 million m³ per year** over the analysed period.

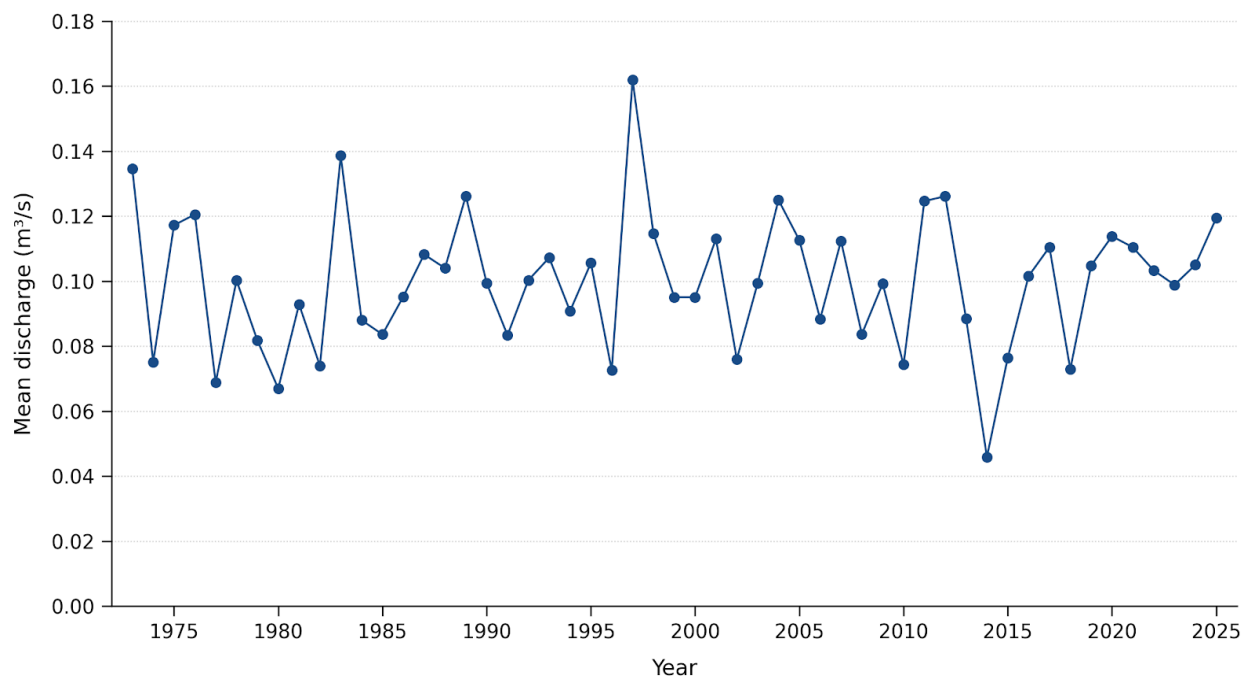


Figure 3. Annual mean discharge at Svarttjønnbekken for 1973–2025.

- Find daily median, 25% percentile og 75% percentile, minimum og maximum for the whole period (use e.g. the PERCENTILE function in Excel). Make a figure that shows the data.

To describe the overall distribution of daily discharge, all daily observations from the 1973–2025 period were combined. The resulting statistics are presented in Table X.

Statistic	Discharge (m³/s)
Minimum	−0.00024

25th percentile	0.0177
Median	0.0448
75th percentile	0.1125
Maximum	2.6981

The median daily discharge is **0.0448 m³/s**, which is considerably lower than the long-term mean discharge of **0.0997 m³/s**. This difference indicates a positively skewed flow distribution, with relatively frequent low-flow conditions interrupted by substantially higher discharge events. Such a distribution is typical of streamflow records, where short-lived high-flow events can contribute disproportionately to the annual runoff volume.

The middle 50% of the observations lies between **0.0177 and 0.1125 m³/s**, as indicated by the 25th and 75th percentiles. The maximum recorded daily discharge is **2.6981 m³/s**, which is approximately two orders of magnitude greater than the lowest typical flows.

The slightly negative minimum value (**-0.00024 m³/s**) is physically negligible and is most likely associated with measurement uncertainty or instrument resolution close to zero flow. The original value was retained in the analysis rather than modifying the dataset.

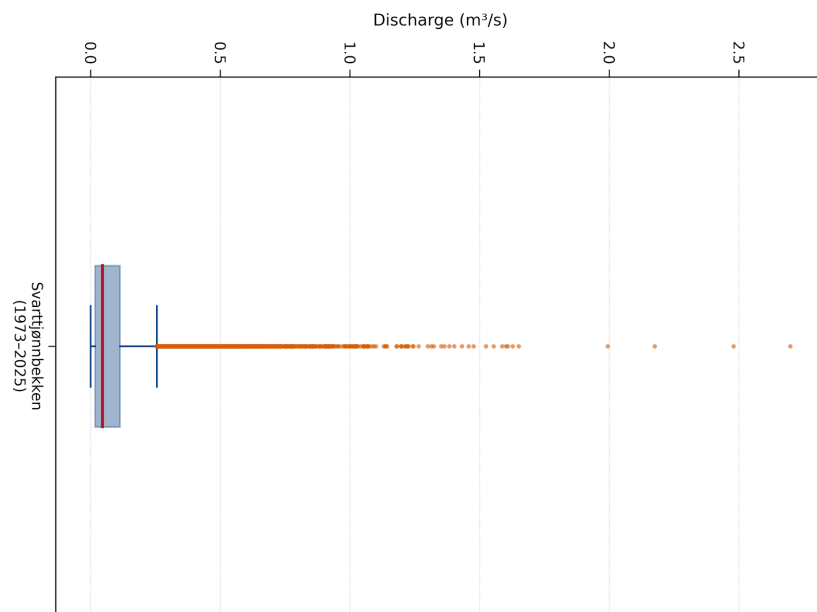


Figure 4. Statistical distribution of daily discharge at Svarttjønnbekken.

- The annual precipitation for Sagelva is 1366 mm. Use this and the result from a) to find the annual evaporation from the catchment.

The annual precipitation for the Sagelva catchment is given as:

$$P=1366 \text{ mm yr}^{-1}$$

The annual runoff depth can be calculated by relating the average annual runoff volume to the catchment area determined in **Task 1a**. If the catchment area is $A \text{ km}^2$, the runoff depth is:

$$R=VA \times 1000106$$

Using the calculated runoff volume:

$$R=3147.8A \text{ mm yr}^{-1}$$

The annual evaporation can then be estimated using a simplified water balance:

$$P=R+E$$

and therefore:

$$E=P-R$$

giving:

$E=1366-3147.8A \text{ mm yr}^{-1}$

The final numerical value of annual evaporation depends on the catchment area obtained from **Task 1a** and will therefore be inserted once the catchment delineation has been completed.

- *Take a look at the plot of annual average discharge found in a), can you see any trend in the data? (Tip: Here you can right click your Excel graph and choose Add Trendline).*

The annual mean discharge was examined for a possible long-term trend. A linear regression was fitted to the annual mean discharge values (Figure X).

The resulting slope is:

$$4.81 \times 10^{-5} \text{ m}^3 \text{ s}^{-1} \text{ yr}^{-1}$$

Although the fitted trend is slightly positive, the coefficient of determination is extremely low:

$$R^2=0.00125$$

This indicates that the linear trend explains virtually none of the observed interannual variability. Consequently, **no clear or statistically meaningful long-term trend can be identified in the annual mean discharge.**

The strong variability between individual years is much more prominent than the very small change represented by the fitted trend. Therefore, based on this dataset, there is no convincing evidence for a systematic increase or decrease in annual mean streamflow at Svarttjønnbekken.

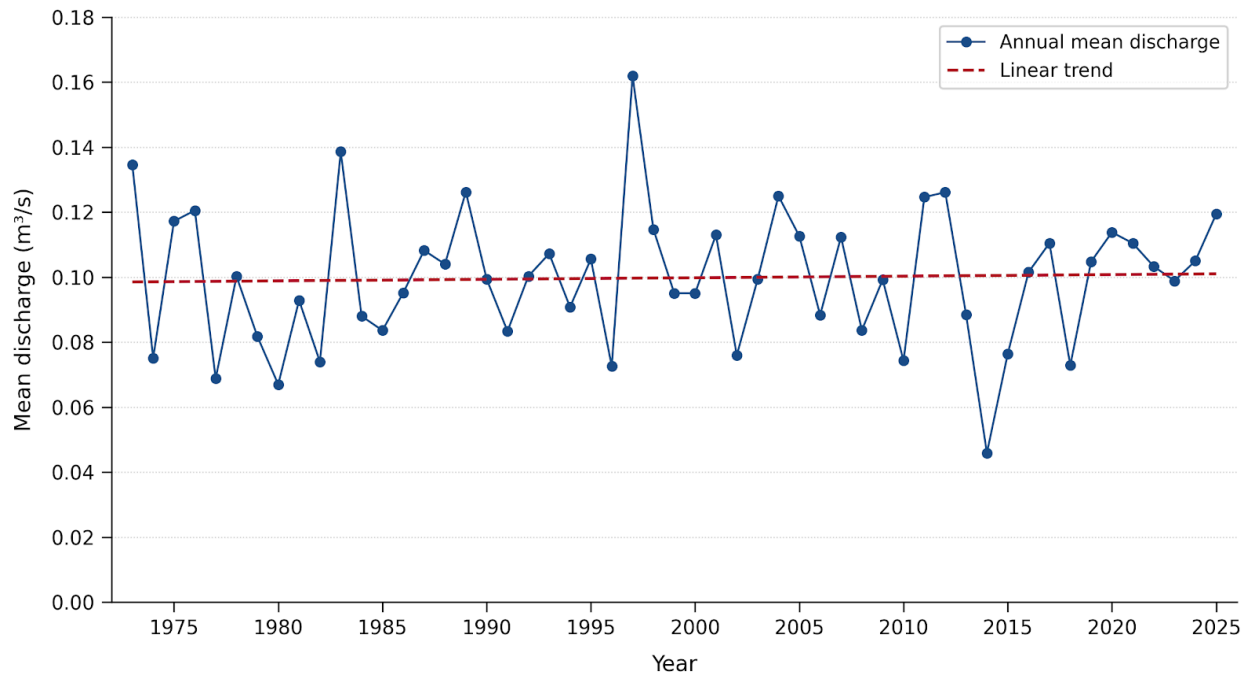


Figure 5. Annual mean discharge and fitted linear trend for 1973–2025.

- Find the annual maximum discharge for each year recorded at the station. Can you see any trends in these?

The maximum daily discharge was identified for each year in order to investigate the variability of high-flow events. The annual maxima shows considerable variability throughout the observation period.

The highest annual maximum recorded in the dataset occurred in **2022**, reaching:

$$Q_{max}=2.698 \text{ m}^3 \text{ s}^{-1}$$

Other notable high-flow years include **2006**, with a maximum discharge of approximately **2.479 m³/s**, **2004**, with **2.175 m³/s**, and **2012**, with **1.994 m³/s**.

A linear regression of annual maximum discharge gives a slope of approximately:

$$0.00104 \text{ m}^3 \text{ s}^{-1} \text{ yr}^{-1}$$

However, the corresponding coefficient of determination is only:

$$R^2=0.00124$$

Therefore, despite the slightly positive fitted slope, the annual maximum discharge does **not show a clear long-term trend**. Instead, the data are characterised by substantial interannual variability and occasional high-discharge events.

From a hydrological perspective, this indicates that individual flood or high-flow years have a much stronger influence on the observed variability than any gradual long-term change apparent in the record.

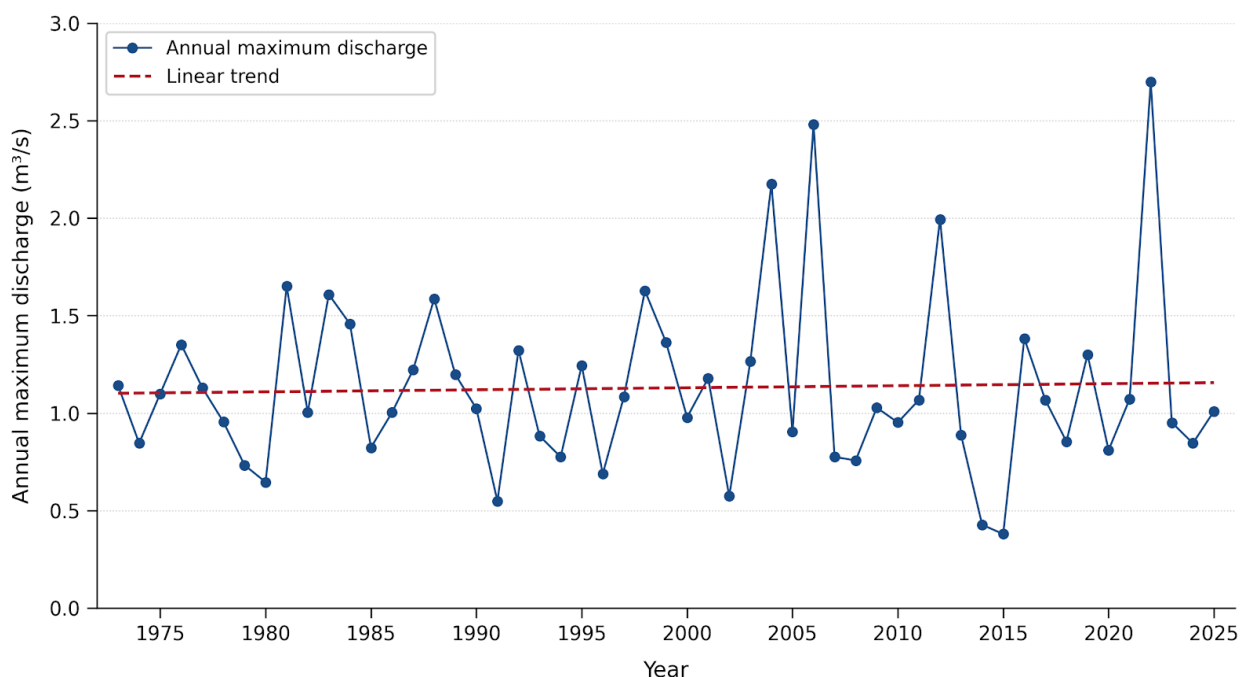


Figure 6. Annual maximum discharge and fitted linear trend for 1973–2025.

Conclusion

The analysis of the Svarttjønnbekken catchment provides a first-order characterisation of the physical and hydrological properties of the basin. The catchment was delineated from the topographic drainage divides and has an estimated area of approximately **3.04 km²**, with elevations ranging from approximately **280 to 513 m a.s.l.** The hypsographic analysis indicates that most of the catchment is concentrated at low to intermediate elevations, while a smaller proportion occupies the higher terrain.

The discharge analysis demonstrates substantial interannual and seasonal variability in the hydrological response of the catchment. The long-term mean discharge provides a representative measure of the average water yield, while the daily discharge statistics illustrate the wide range of flow conditions experienced at the gauging station. The comparison between precipitation and calculated runoff further provides an approximate water balance for the catchment and highlights the proportion of precipitation that is not accounted for as streamflow.

The annual mean and annual maximum discharge series do not indicate a pronounced long-term trend over the analysed period; instead, the data are characterised primarily by considerable year-to-year variability. This variability is likely related to changes in precipitation, snow accumulation and melt, and other meteorological and catchment processes. The topographic setting of the basin is particularly relevant in this respect, as the elevation range can influence the partitioning and timing of precipitation and runoff.

Overall, the exercise demonstrates how topography, catchment geometry and long-term discharge observations can be combined to obtain an initial understanding of a catchment's hydrological behaviour. The resulting characterisation provides an appropriate basis for more detailed hydrological analyses of the Svarttjønnbekken catchment later in the course.